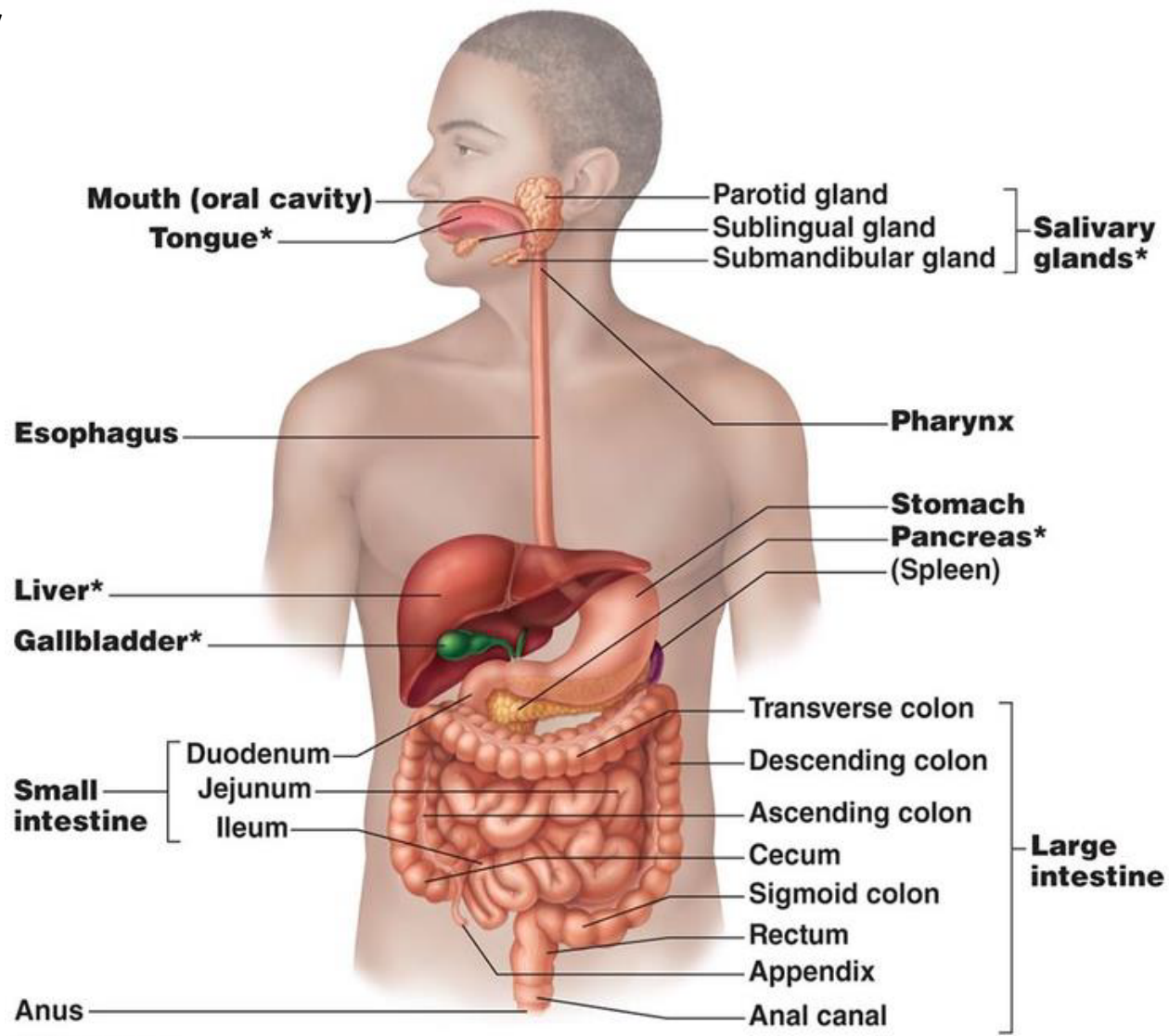


Immunological aspect of the gastrointestinal tract

Overview

The gastrointestinal tract (GIT) is a **critical interface between the external environment and the internal milieu**, working not only digestion, secretion, motility, and nutrient absorption, but also as a **gateway** for potential **harmful environmental elements** including: **pathogens, environmental toxins or dietary Ags.**



The gut immune system constantly balances:

- **Inflammation** (when needed) vs. **Suppression** (to prevent collateral damage)
- **Active immunity** vs. **Tolerance**
- **Protection** vs. **Autoimmunity/Allergy**
- Failure in any of these balances leads to clinically significant disease.
-

The Intestinal Immune System

- The intestinal immune system is predominantly composed of:
 - The gut microbiota
 - Mesenteric lymph nodes
 - Specialized epithelial cells
 - Innate and adaptive immune cells
 - Related metabolites

The Intestinal Immune System

- **The Microbial Population (The Microbiota):** The GIT harbors millions of microorganisms (bacteria, fungi, viruses) that form a complex ecosystem living in symbiosis with the host.
- **The Danger of Imbalance (Dysbiosis):** When the normal microbial composition is disrupted—whether by antibiotics, poor diet, or infection—it results in **dysbiosis**.
- **Dysbiosis** is **linked to various diseases**, such as anxiety, depression, hypertension, cardiovascular diseases, obesity, diabetes, inflammatory bowel disease, and cancer.

The main functions of the human intestinal microbiome include:

- (i) Protection against pathogenic microbial colonization.
- (ii) Maintenance of the barrier function of the intestinal epithelium.
- (iii) Degradation and absorption of nutrients.
- (iv) Active regulation of the intestinal immunity.

The immune system plays a key role in GIT pathologies

- ✓ being responsible for protecting the body against infection
- ✓ maintaining homeostasis,
- ✓ and regulating the inflammatory response.

The improper functioning of the immune system is a critical element in **GI diseases** includes:

- ✓ inflammatory bowel disease (IBD),
- ✓ irritable bowel syndrome (IBS),
- ✓ food allergy,
- ✓ and GI infections triggered by pathogens

- ✓ **Innate immunity:**
- ✓ provides the first line of defense mechanisms eg. :
 - ✓ mucosal barrier ,
 - ✓ effector cells,
 - ✓ antimicrobial peptides,
 - ✓ soluble mediators, and cell receptors.
- ✓ **Adaptive immunity**, involving **B and T lymphocytes**,
- ✓ provides **long-term immunity** through **immunological memory**, ensuring a precise response to the repeated exposure to pathogens.
- ✓ **Understanding of the interactions between the immune system and the GIT** is essential for the development of **new therapies** and **management strategies** to **improve digestive health**.

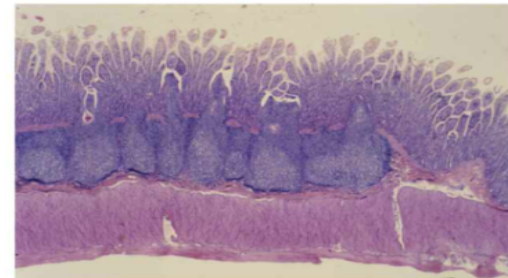
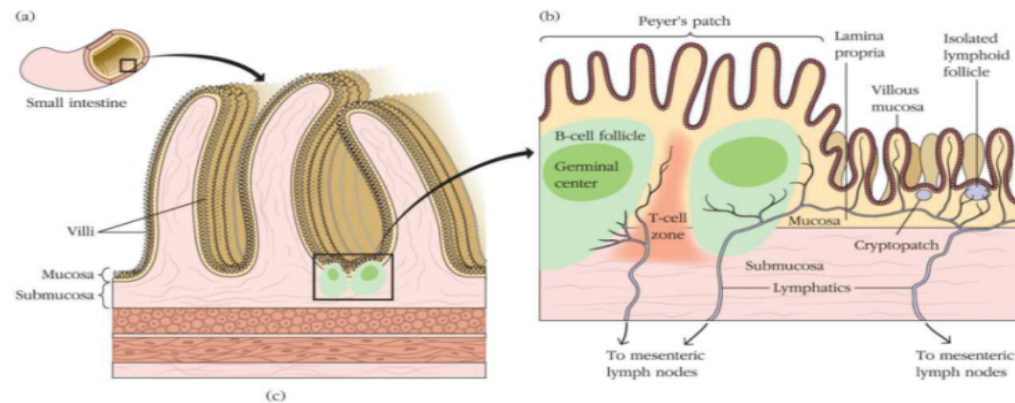
Examples of innate defense system of GIT

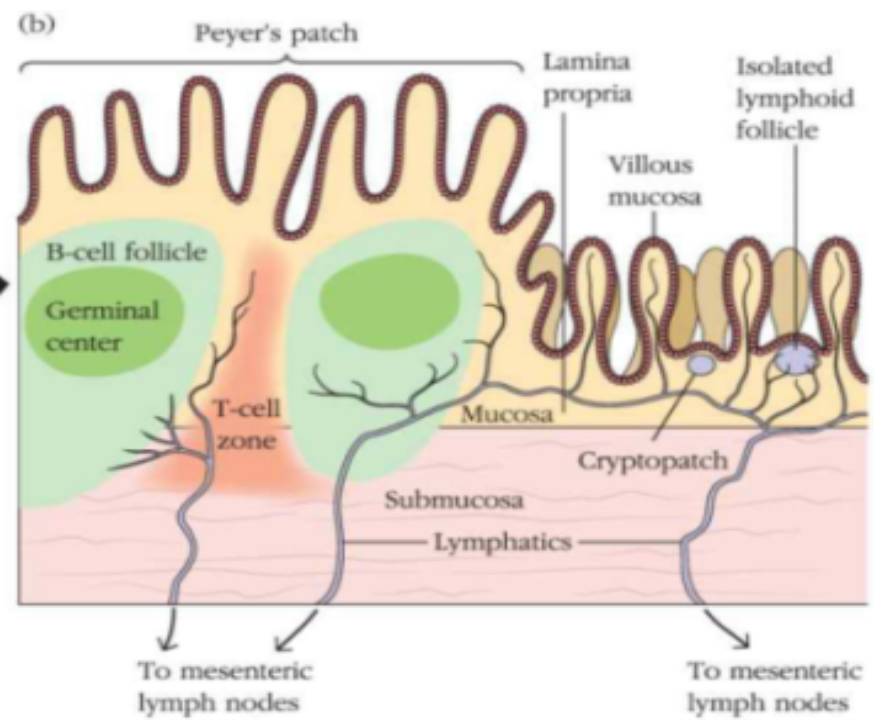
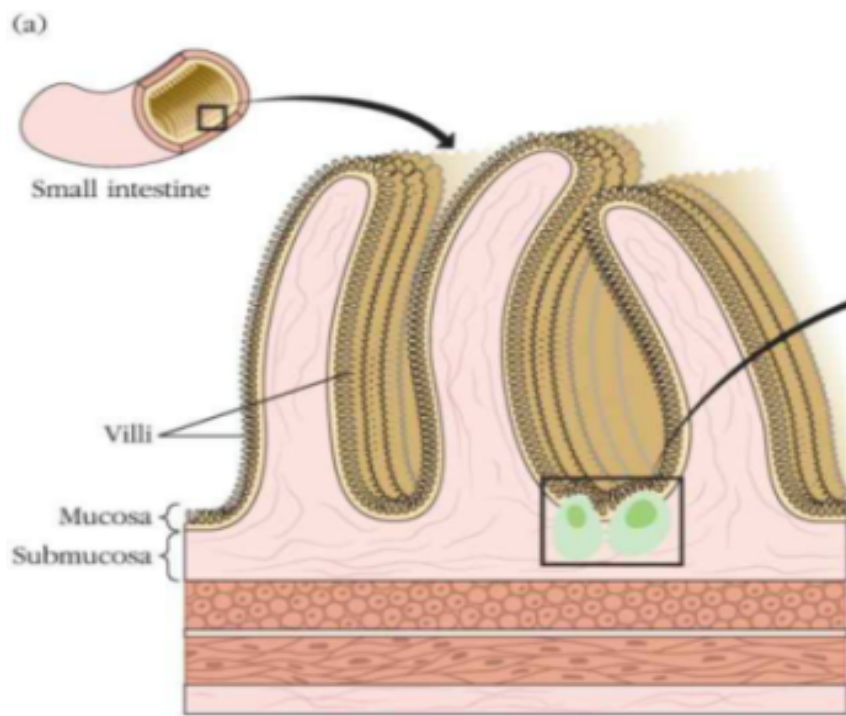
- The tight junction between the epithelial cells.
- Salivary defense proteins (eg. lysozyme) and pancreatic antimicrobial peptides (AMPs).
- The acidity of the stomach.
- Secreting mucus and antimicrobial factors produced by the intestinal mucosa.
- Bile acids as immune modulators by acting as signaling molecules that bind to specific receptors on macrophages and DCs to maintain the tolerogenic state of the liver and intestine immunity.
- Microbiome and microbiota-derived products or their metabolites.
- Normal epithelial cell turnover and intestinal motility act together to purge the intestinal tract of harmful micro-organisms.

- **Changes** in these factors can lead to ↑ susceptibility to infections and various diseases.
- **Many organisms and bacterial overgrowth** is a consequent complication in patients with achlorhydria due to **atrophic gastritis**.
- **Any change in the normal microflora** of the intestine allows pathogenic bacteria to flourish: **pseudomembranous colitis** caused by the toxin-producing bacterium *Clostridium difficile* in patients given certain antibiotics.
- **Disordered gut motility** leads to small intestinal bacterial overgrowth, ↑ mucosal permeability, and bacterial translocation.
- **Disturbances in intestinal tight junctions** lead to ↑ intestinal permeability (leaky gut) which allows luminal toxins, bacteria, and Ags to enter the bloodstream and tissues, triggering systemic inflammation and contributing to various local and systemic diseases like **IBD**.
- **Imbalances in bile acid levels and signaling pathways** are implicated in the development and worsening of conditions like **Crohn's disease and ulcerative colitis disease**.

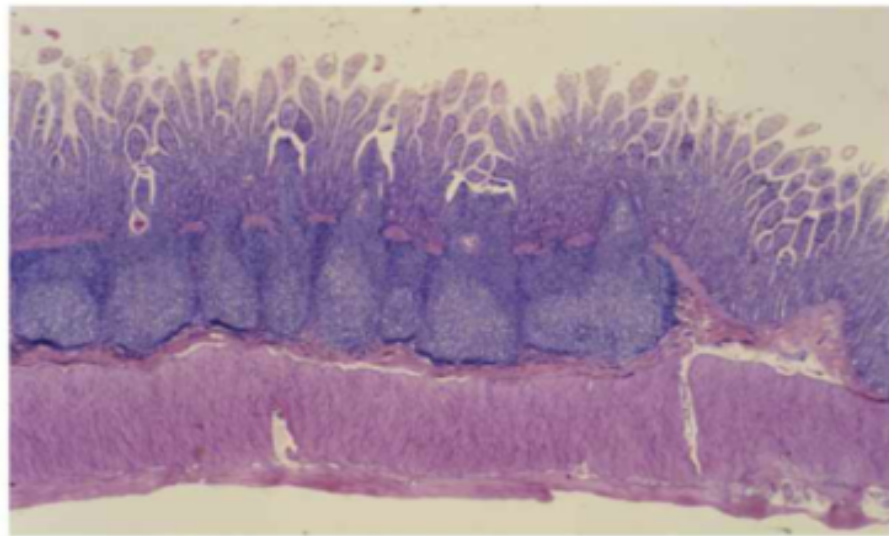
Gut-Associated Lymphoid Tissue (GALT)

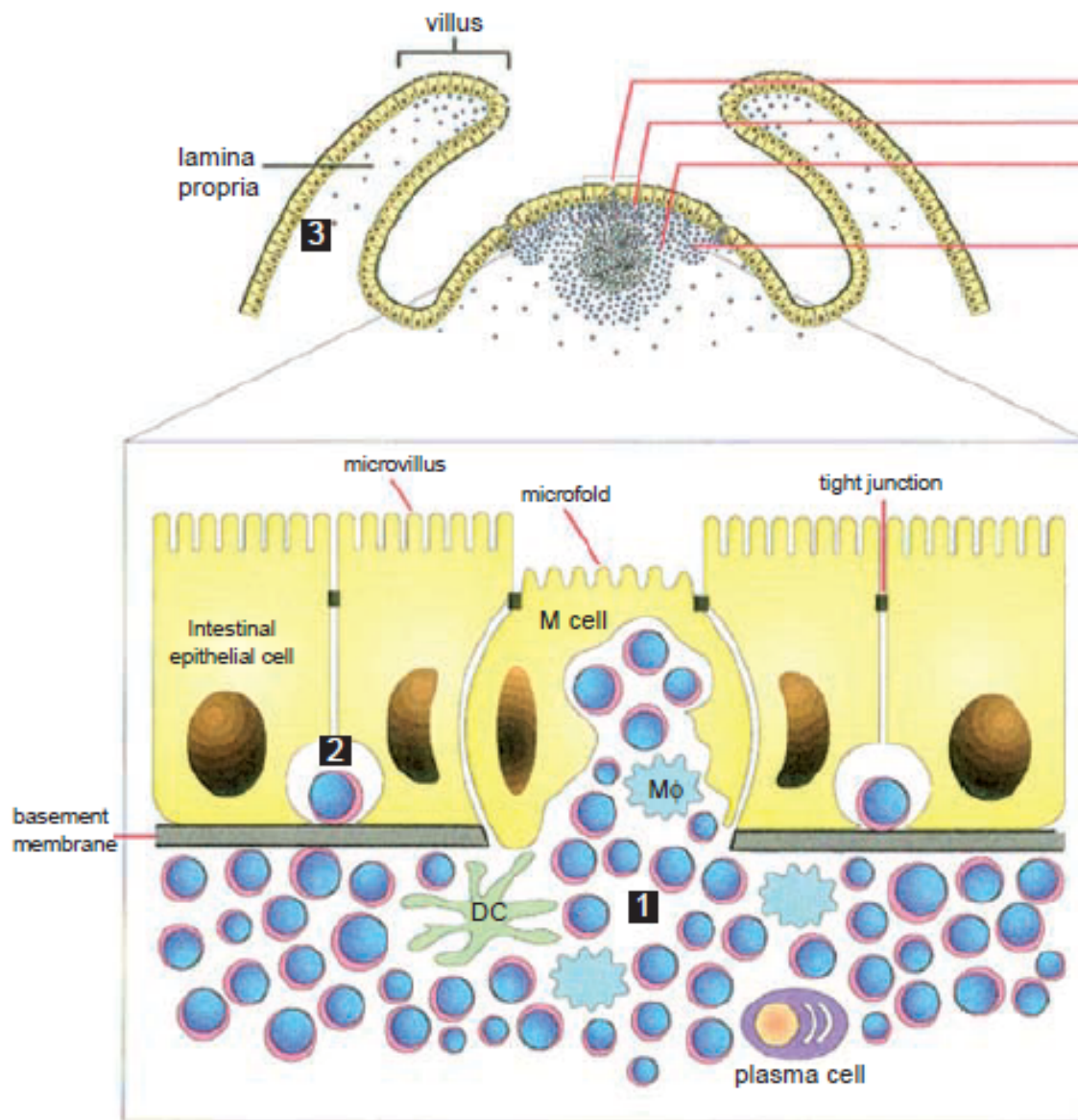
- ✓ Are the key Ag sampling and adaptive immune inductive sites within the intestine wall which monitor and prevent the growth of pathogenic microbes in the GIT, it includes:-
 - **Peyer's patches (PP)** are lymphoid follicles situated along the wall of the small intestine (their density ↑ towards the terminal ileum) that trap Ags from the GIT and provide sites where B and T cells can interact with Ag.





(c)



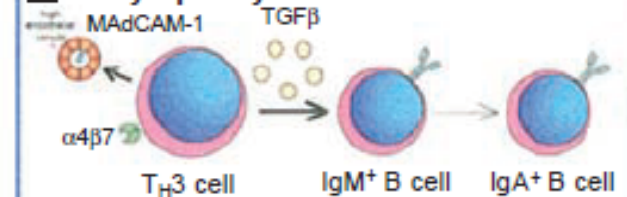


COMPONENTS OF THE PEYER'S PATCH

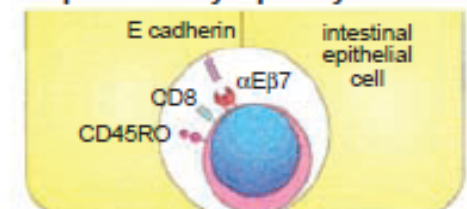
- Follicle-associated epithelium
- Dome region
- Follicles - B cell zones with germinal centers
- Parafollicular regions - enriched with T cells

LYMPHOCYTES OF THE GALT

1 PP lymphocytes



2 Intraepithelial Lymphocytes



3 Lamina Propria Lymphocytes

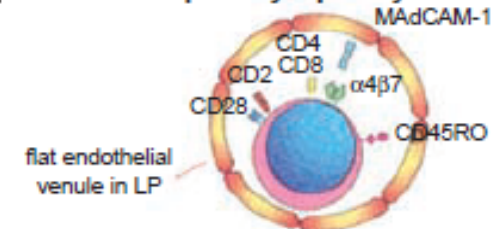


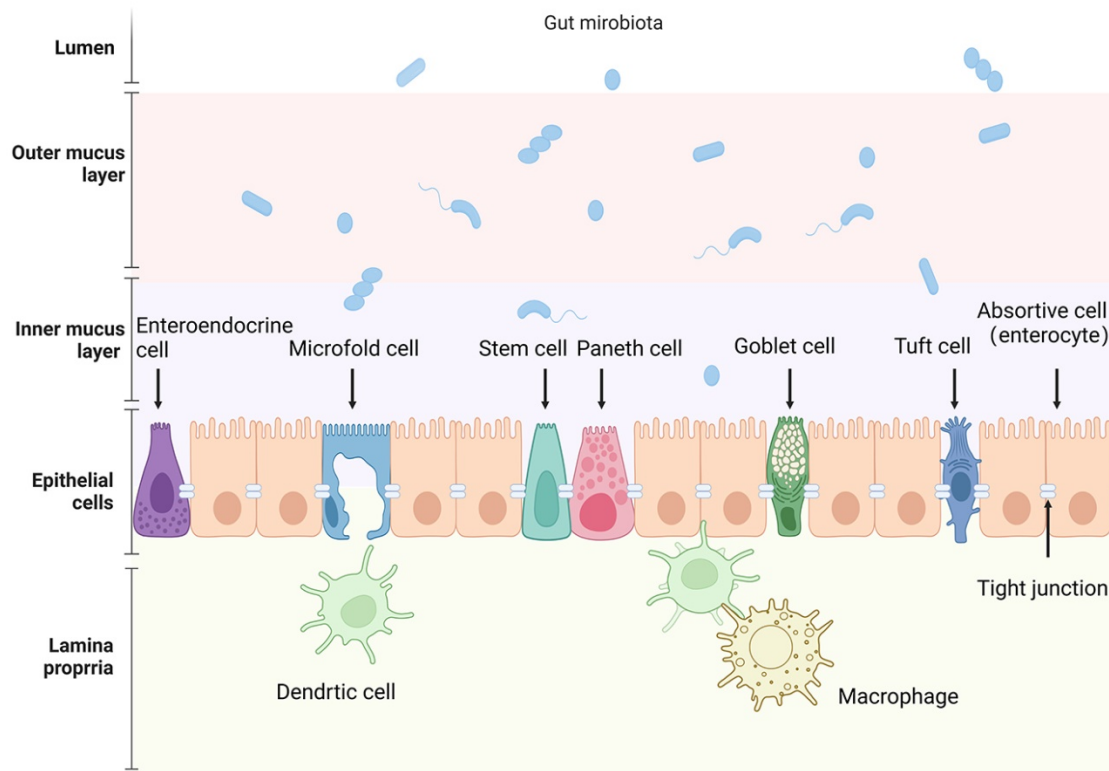
Figure 14.1 Anatomy of the gut-associated lymphoid tissue.

- **Vermiform appendix** represents a *bona fide* adaptive immune inductive site with a similar structure and immune cell composition to PP.
- **Isolated lymphoid follicles (ILFs)**, which are associated with epithelial surfaces along the entire intestinal tract.
 - The locale and size of ILFs change relatively quickly in response to changes in the intestinal microbiome and cytokine milieu.
 - Their cellular composition, density and location within the gut wall varies between intestinal segments.
 - They are a primary source of IgA-secreting B cells.

The **epithelial cells of intestine** are diverse in phenotype and function and are now recognized as a **key part of innate immune system**:-

- ✓ **Enterocytes (Absorptive cells)**, do express TLR4 internally.....etc.
- ✓ **Enteroendocrine cells....** key sensors of microbial metabolites, can release cytokines in response to pathogen associated molecules and secrete hormones (eg. serotonin) regulating gut physiology.
- ✓ **Goblet cells**, produce mucins to form a protective mucus layer and they secrete AMPs that inhibit the activity of luminal microbes that come too close to their membranes.
- ✓ **Paneth cells**, secrete a variety of AMPs (eg. α -defensins, lysozyme...etc.) that protect the gut epithelium.

- ✓ **Microfold (M) cells** are highly specialized for the transcytosis of Ag across the epithelium and interact with hemato-poietic cells including macrophages, DCs, and lymphocytes.
- ✓ **Tuft cells** detect helminthes and initiate type 2 immune response.



- Defects in enterocyte structure or differentiation is associated with congenital enteropathies.
- Abnormalities in Paneth cells are associated with many human disease processes, including Crohn's disease.

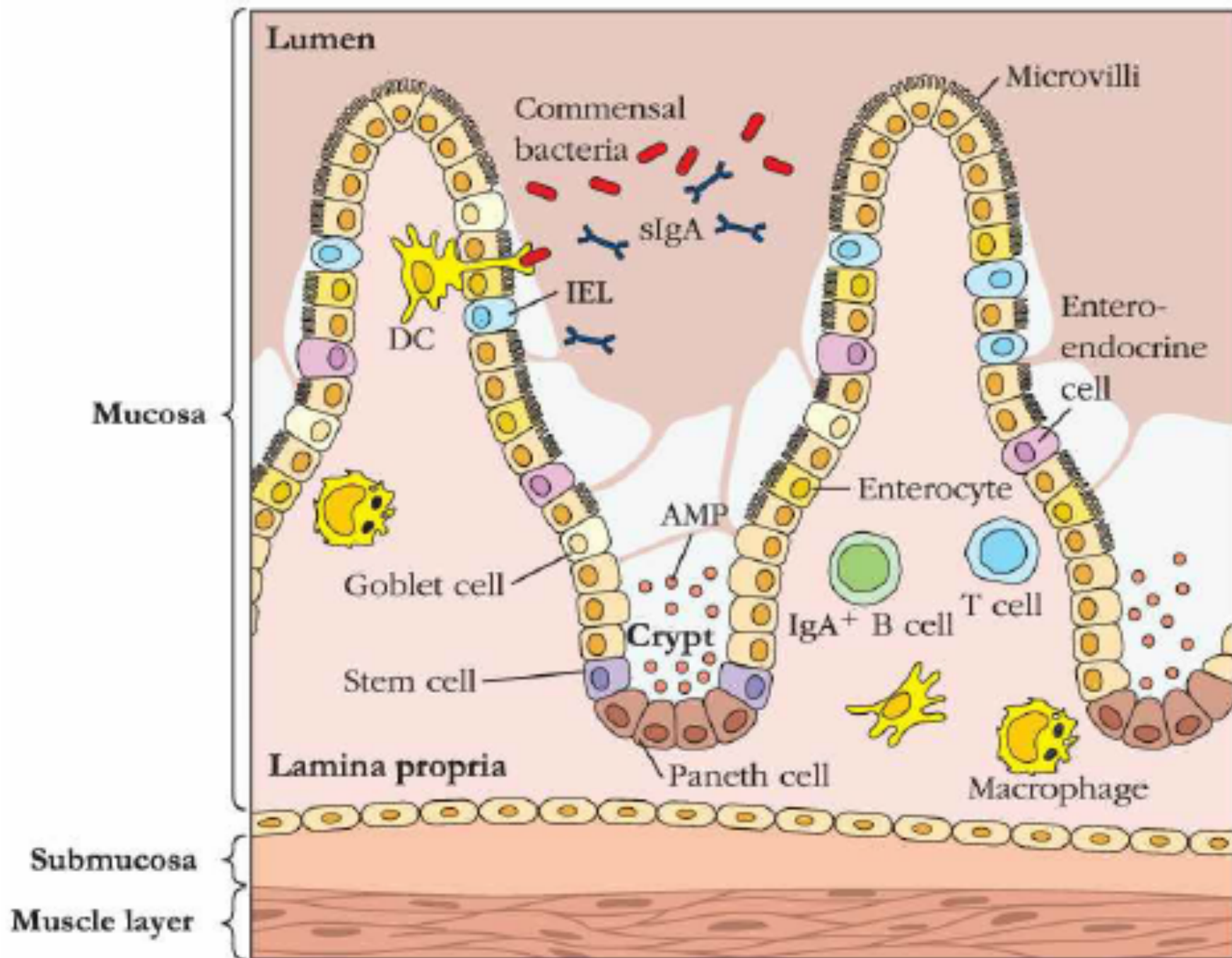
❑ The **lamina propria (LP)** contain most of the immune cells involved in **maintaining tolerance and/or mounting an inflammatory response.**

- Resident and migratory APCs and mast cells are predominant
- Innate lymphoid cells (ILCs).
- T cells (CD+4, CD+8, Th17 cells and Treg cells)
- B cells.....(plasma cells in the healthy

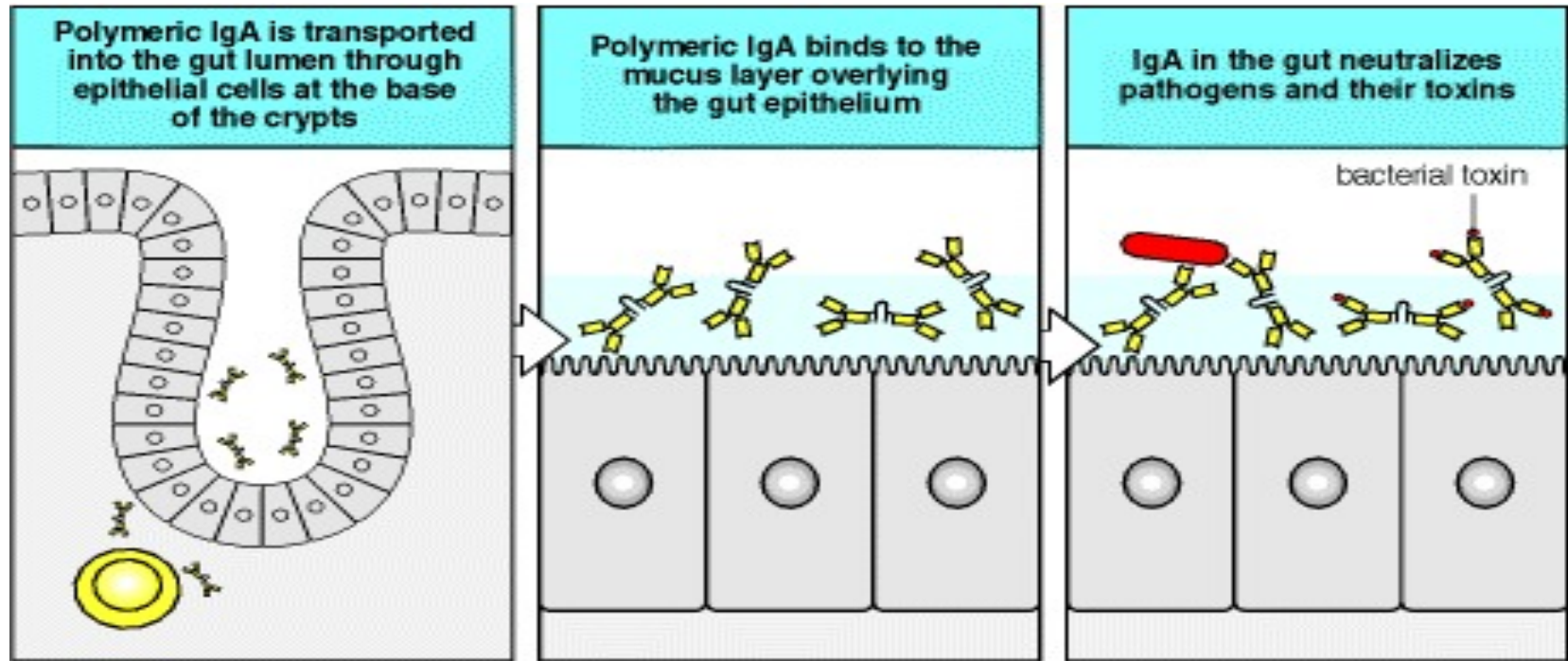
lamina propria secrete large amounts of IgA)

❑ During an inflammatory immune response, other types, including granulocytes, are recruited to the barrier tissue.

- **Th1 cells:** Cell mediated responses (intracellular pathogens).
- **Th2 cells:** Ab mediated responses (allergens, parasites).



Secretory IgA the most abundant class of Ab found in the intestinal lumen of humans associated with the mucosal immune system.



IgA is synthesized by plasma cells in the lamina propria and transported into the lumen of the gut through epithelial cells at the base of the crypts.

IgA exerts its gut immune effect through:-

- Neutralization of the toxin (cholera toxin), virus and bacteria.
- Blocks the adherence of pathogenic microorganism to gut epithelium.
- Blocks the uptake of antigen into systemic immune system.

- The homeostatic environment of the gut is dominated by anti-inflammatory cytokines including IL-10, and TGF- β .
- Cells that maintain barrier health, including T cells, IgA-producing B cells, and T17 cells, and intraepithelial lymphocytes.
- APCs in the healthy intestine sample Ag, travel to secondary lymphoid tissue, and polarize naïve T cells to the T lineage (effector cells).
- IgA-producing B cells are generated in both T cell–dependent and T cell–independent manners.
- ILC3 and Th17 cells are also abundant in the intestinal LP and play dual roles. They help maintain homeostasis by producing IL-22, which stimulates epithelial cells to produce protective AMPs. However, they can also contribute to inflammatory responses.
- Eosinophils are responsible for essential effector functions during parasitic infection and allergic responses, but they also contribute to normal gut homeostasis.

Food Intolerance

- **A non-immune-mediated reactions**, in which the human body cannot properly digest or metabolize certain food components leading to GI discomfort and other non-immune-related symptoms.
- **These reactions** can be enzymatic, pharmacological or metabolic.
- **EXAMPLES**....Lactose intolerance and Fructose intolerance and histamin intolerance...etc
- **Lactose Intolerance**:- Patient does not have the enzyme lactase and, therefore, cannot break down sugar lactose in milk and other foods into glucose and galactose.
- **Profound diarrhea or vomiting** usually results without any other system involved (eg. **respiratory or skin**).
- **Food avoidance** is the **treatment** of choice.

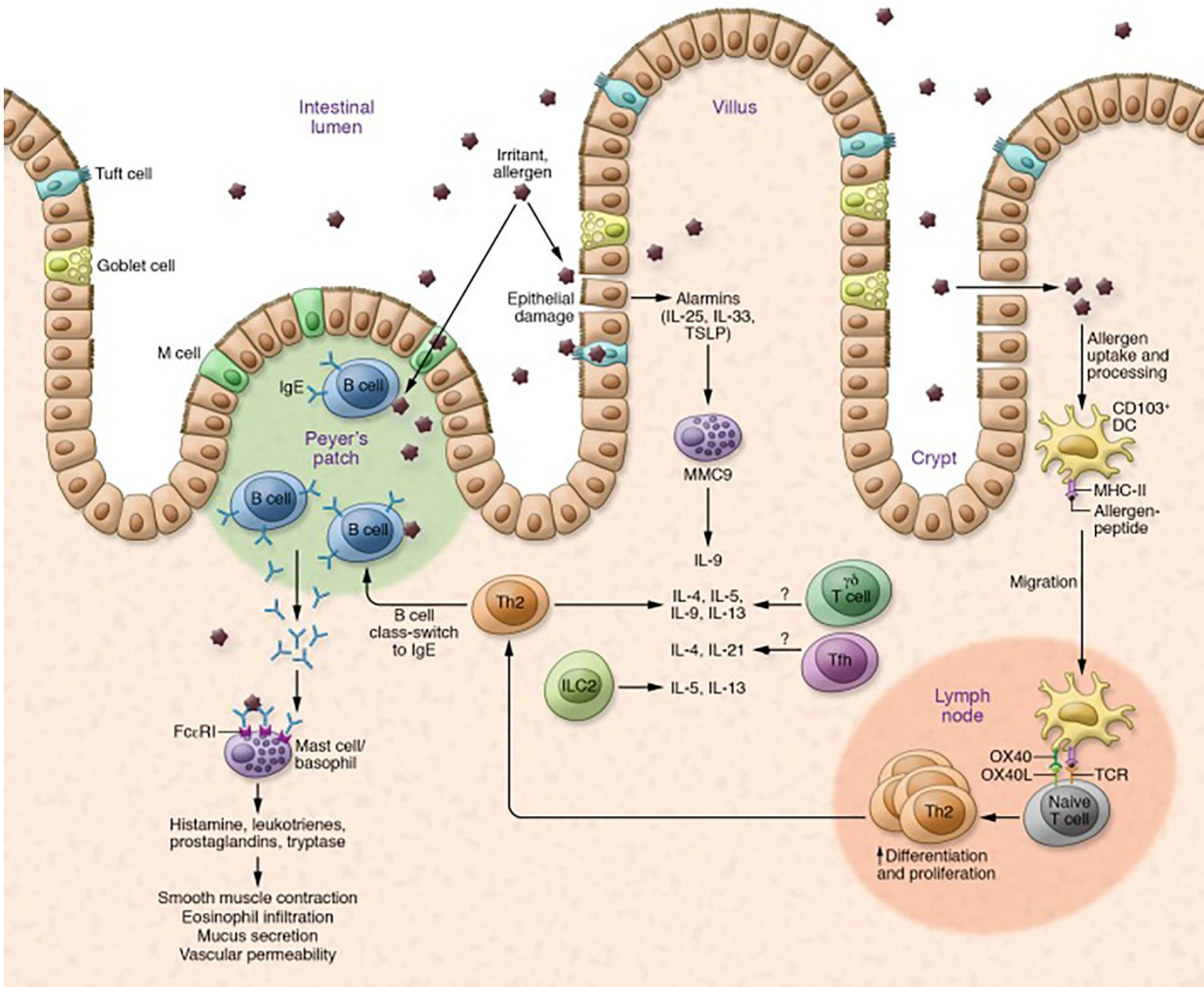
Food Allergy

- **Immune-mediated disorders**, They can be IgE-mediated (Th2 inflammatory response), non-IgE mediated, or mixed.
- The most common **culprits** are peanuts, milk, eggs, peanuts, fish....etc.
- The **pathophysiological cascade** initiates during the sensitization phase when genetically predisposed individuals encounter these **food Ags**, triggering a complex immunological cascade that manifests as multi-systemic inflammation affecting skin, GI mucosa, and respiratory system.
- **Clinical manifestations** occur immediately on exposure and range from **minor** to **life-threatening** or even fatal **anaphylaxis**.
- The **symptoms of immediate GI** hypersensitivity include nausea, abdominal pain, cramps, vomiting, and/or watery/mucous diarrhea.
- The **diagnosis is difficult** to prove or refute and skin-prick tests have limited predictive value.

- In many cases, clinical suspicion and trials of elimination diets are used.
- Treatment of proven food allergy consists of detailed patient education, strict elimination of the offending Ag.
- In some cases, antihistamines and immunotherapy (eg. Subcutaneous immunotherapy for desensitization in patients with peanut allergy).
- Anaphylaxis should be treated as a medical emergency.

Mechanism of allergic reaction in FA. Allergic reaction is mediated by upregulation of Th2 cytokines and B cell class switching, which leads to production of IgE antibodies and degranulation of mast cells and basophils. Degranulation initiates allergic responses, such as smooth muscle contraction, eosinophil infiltration, mucus secretion, and vascular permeability. Other cells such as Tfh cells, $\gamma\delta$ T cells, and ILC2s are also thought to play a role in allergic reactions. Key cytokines involved are IL-25, IL-33, IL-4, IL-5, IL-13, and IL-9.

Adapted from: Sampath V, Nadeau KC. Newly identified T cell subsets in mechanistic studies of food immunotherapy. *J Clin Invest.* 2019;129(4):1431-1440. doi:10.1172/JCI124605.



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